Rigour, Relevance and Relationships

Keeping a research career alive while making a difference in policy

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Executive Director, Motu Economic and Public Policy Research · Chair, IRANZ

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The question you asked me to answer

“How do you keep an active research career while making a difference in policy?”



ACT 1

Foundations

Yale PhD and the econometrics that underpin everything after



ACT 2

Policy institutions

IMF and Reserve Bank: technical depth, and the co-authors who kept it current



ACT 3

Leading research

Motu and IRANZ: building the conditions for other people’s research

Four decades, one thread

1986–88
BCom/BSc, MCom
University of Auckland

1988–90
Economist
Reserve Bank of NZ

1990–94
PhD, Yale University
Cowles Foundation

1994–98
Economist
IMF, Washington DC

1998–2005
RBNZ, then Chief Economist
National Bank

2006–07
Associate Professor
Victoria University

2007–19
Assistant Governor
Reserve Bank of NZ

2019–
Executive Director, Motu
Chair, IRANZ (2021–)



Academic / research

Policy institutions

Research leadership

ACT 1

The technical foundation

Yale, the Cowles Foundation, and learning to prove that the tools we use actually work

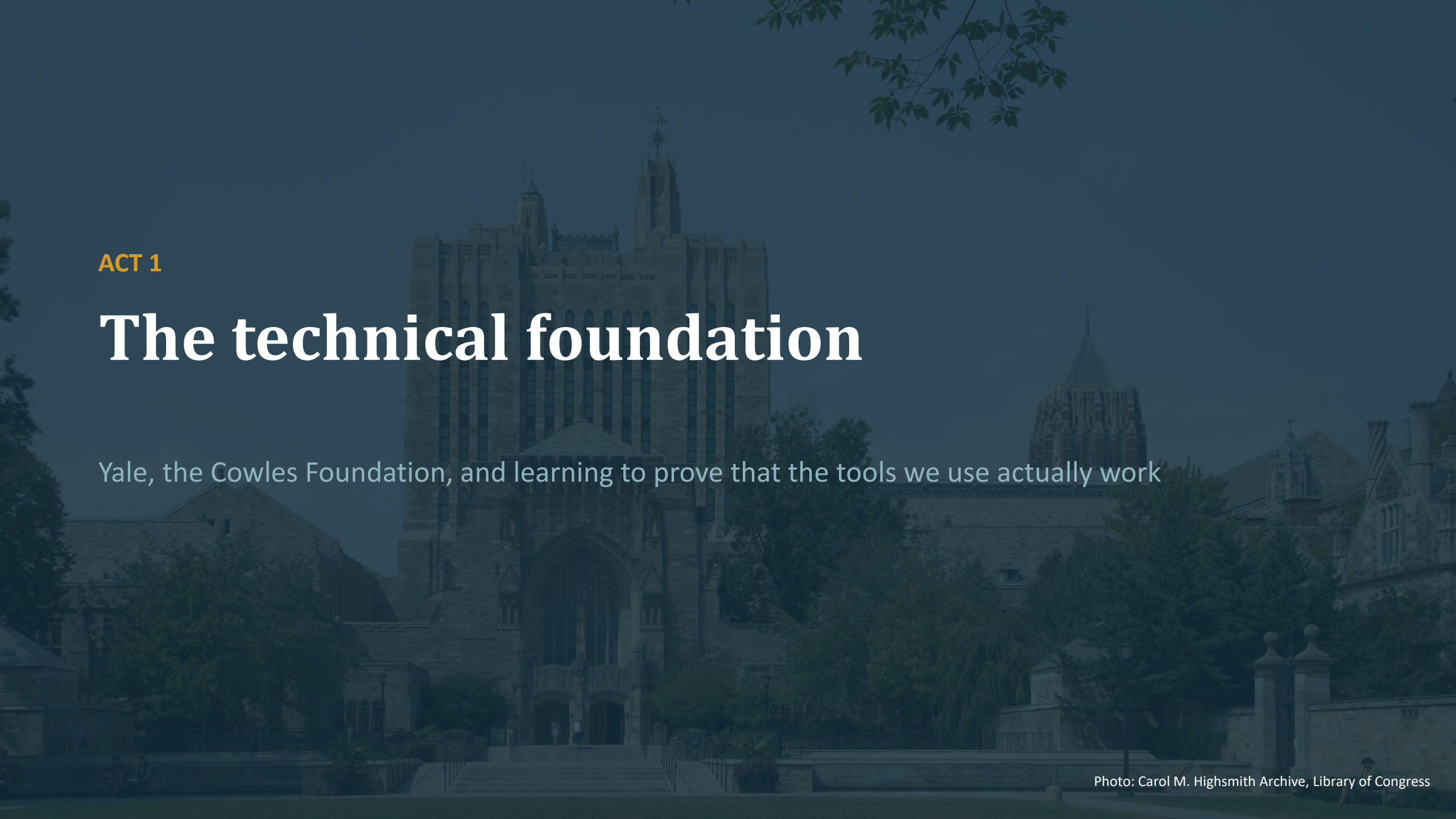


Photo: Carol M. Highsmith Archive, Library of Congress

Yale: training at the frontier

1994

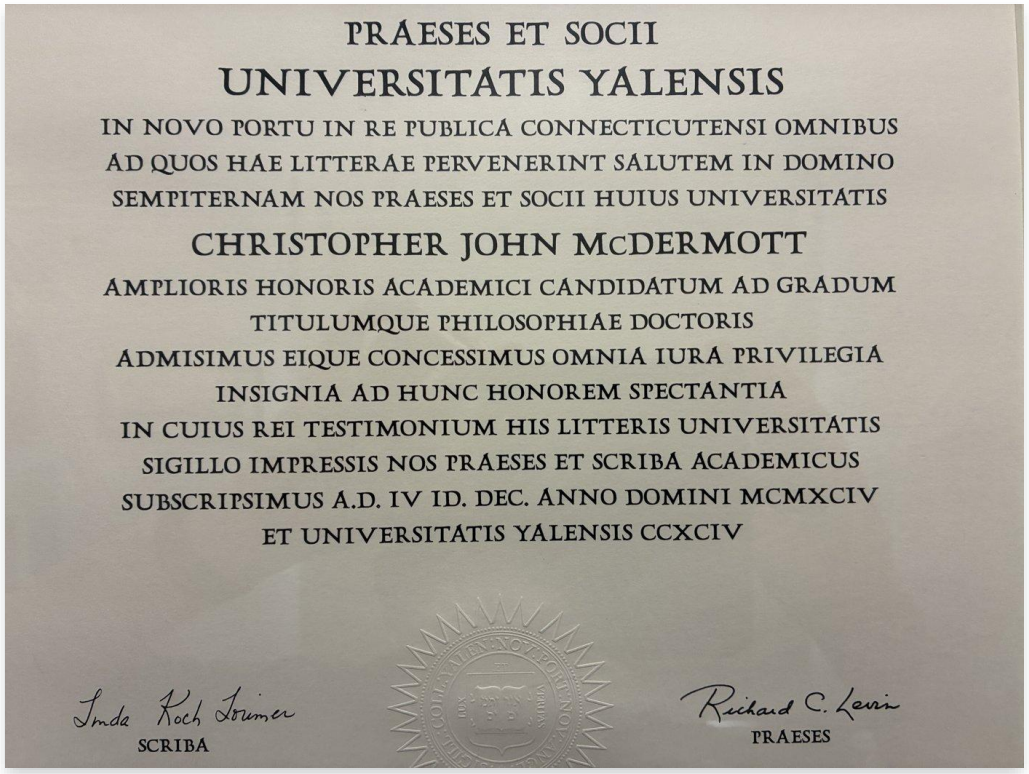
PhD, Yale University
Conferred Dec 1994, presented May 1995

Sloan

Dissertation Fellowship
Alfred P. Sloan Foundation, 1993/94

Bergstrom

Prize in Econometrics
A.R. Bergstrom Prize, 1995


PRAESES ET SOCII
UNIVERSITATIS YALENSIS
IN NOVO PORTU IN RE PUBLICA CONNECTICUTENSI OMNIBUS
AD QUOS HAE LITTERAE PERVENERINT SALUTEM IN DOMINO
SEMPITERNAM NOS PRAESES ET SOCII HUIUS UNIVERSITATIS
CHRISTOPHER JOHN McDERMOTT
AMPLIORIS HONORIS ACADEMICI CANDIDATUM AD GRADUM
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ET UNIVERSITATIS YALENSIS CCXCIV
Linda Koch Lorimer
SCRIBA
Richard C. Levin
PRAESES

Before Yale: Auckland MCom (first-class honours) and two years at the RBNZ. **At Yale:** co-authored with Donald W. K. Andrews, one of the leading econometric theorists of his generation, published in the *Review of Economic Studies* (1995).

The problem: trends in nonlinear models

Andrews and McDermott (1995), Review of Economic Studies 62, 343–360

Observed data = trend ∘ stochastic part

$$W_t = d(t, Z_t), \quad t = 1, \ldots, T^*$$

Trend shapes economists actually use:

$d(t) = bt + ct^2$

$d(t) = exp(bt)$

$d(t) = bt \text{ for } t \le G, \; ct \text{ for } t > G$

Polynomial · exponential (levels) · broken trend

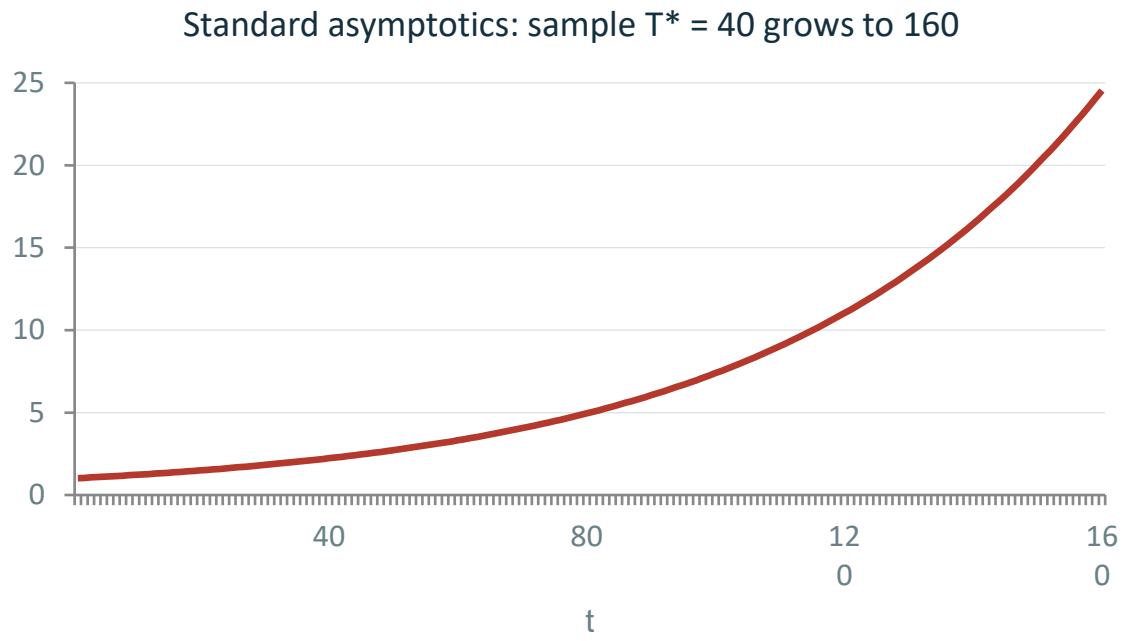
Why standard asymptotics struggle

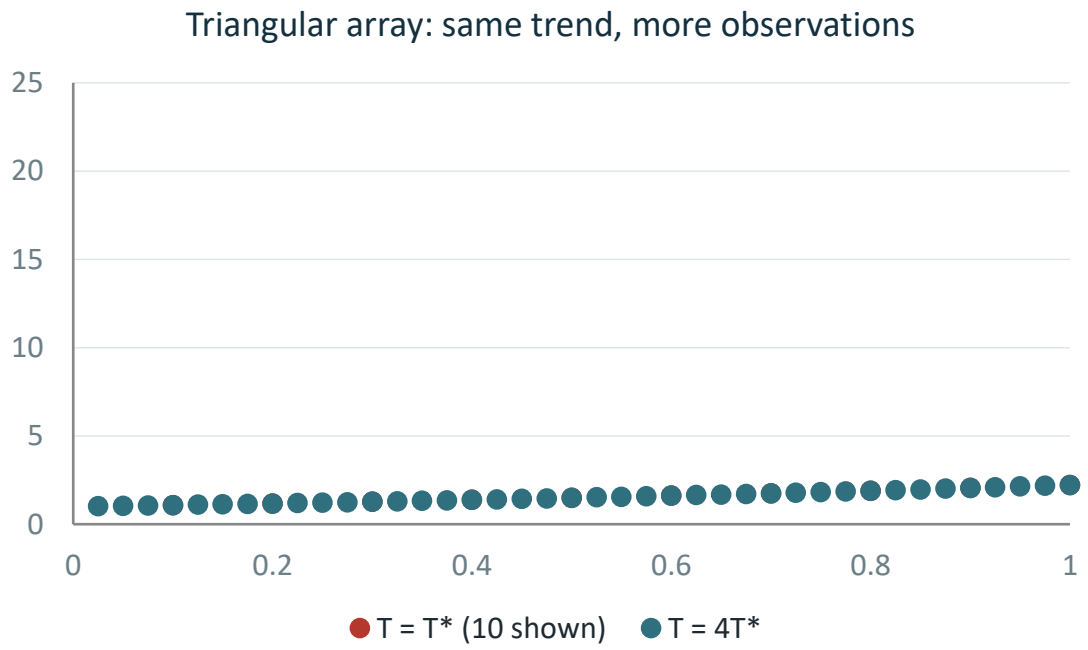
- As T → ∞ the trend's magnitude explodes, so the asymptotics stop mimicking the sample we actually have
- With an exponential trend in levels, the Lindeberg condition fails: no CLT, even with iid bounded shocks
- Results depend on the trend for t far beyond the sample, which is irrelevant to the data in hand
- Existing results (sequential asymptotics) rest on high-level conditions that are hard to verify

The idea: a triangular array

$$W_{Tt} = d(\,(T^*/T)\cdot t\,,\, Z_t\,),\quad t = 1, \ldots, T,\quad T \geq 1$$

At T = T* this is exactly the observed sample. As T grows we observe the same trend shape more densely, not a longer trend.





Illustrative: exponential trend with b = 0.02, T = 40*

The result: the standard toolkit is justified

For GMM / extremum estimators under the triangular array

$$\sqrt{T}\,(\theta_T - \theta_0) \to^d N(0, V)$$

θ_T: GMM estimator from a sample of size T

Wald, LM and LR-type test statistics $\to \chi^2$ *under the null*

- V depends on the form of the trend…
- …but the usual covariance estimators designed for non-trending data are still consistent
- So estimation and inference proceed exactly as in the non-trending case



Why it mattered to me

It taught me to ask whether an approximation fits the problem actually in front of you. That question turned out to be the most useful thing I took into policy work.

ACT 2

Research inside policy institutions

IMF 1994–98 and Reserve Bank of New Zealand: why technical depth pays, and how co-authors kept it current

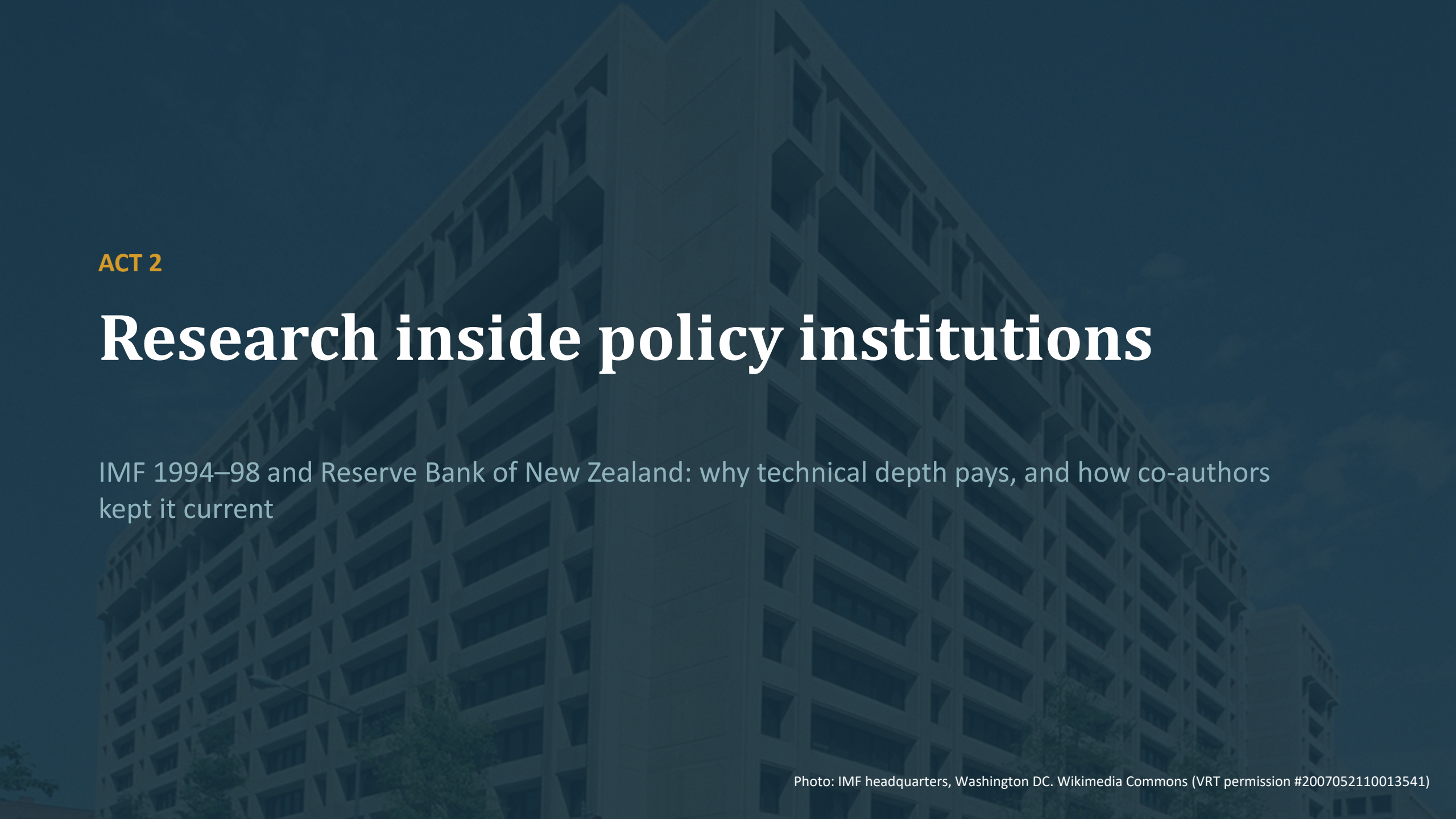


Photo: IMF headquarters, Washington DC. Wikimedia Commons (VRT permission #2007052110013541)

Policy questions are econometric questions in disguise



Is this shock temporary or permanent?

Persistence decides whether to finance or adjust: smooth through it, or change course.



Is the trend real, or just noise?

A small trend can be swamped by variability. Policy should respond to what dominates.



Has the economy changed?

Structural change shifts the transmission of policy. Old rules of thumb can mislead.

Paul Cashin and the IMF: commodity prices

13

journal articles
1998–2006

140

years of real commodity prices, 1862–1999

~1%

a year: the small long-run decline in real prices

Small trends, big variability

Real prices drifted down about 1% a year, but variability dwarfs the trend and rose after Bretton Woods. The trend has little practical policy relevance.

Slumps outlast booms

Price slumps last longer than booms, and the chance a slump ends does not depend on how long it has lasted.

Shocks are persistent

Shocks are typically long-lived, so stabilisation schemes for export earnings are likely to cost more than they deliver.

Real exchange rates

Parity reversion in real exchange rates is faster for developed countries and flexible regimes. Terms-of-trade shocks shape saving and current accounts.

Viv Hall and Victoria: NZ's business cycle

2011

Build the data

No official quarterly real GDP before 1977. We built a seasonally adjusted quarterly series from mid-1947 (Chow–Lin temporal disaggregation).

2016

Date the cycle

Bry–Boschan turning points: nine recessions in ~70 years. GDP and employment cycles coincide ~90% of the time, and recovery strength is independent of recession depth.

2004–12

Test policy options

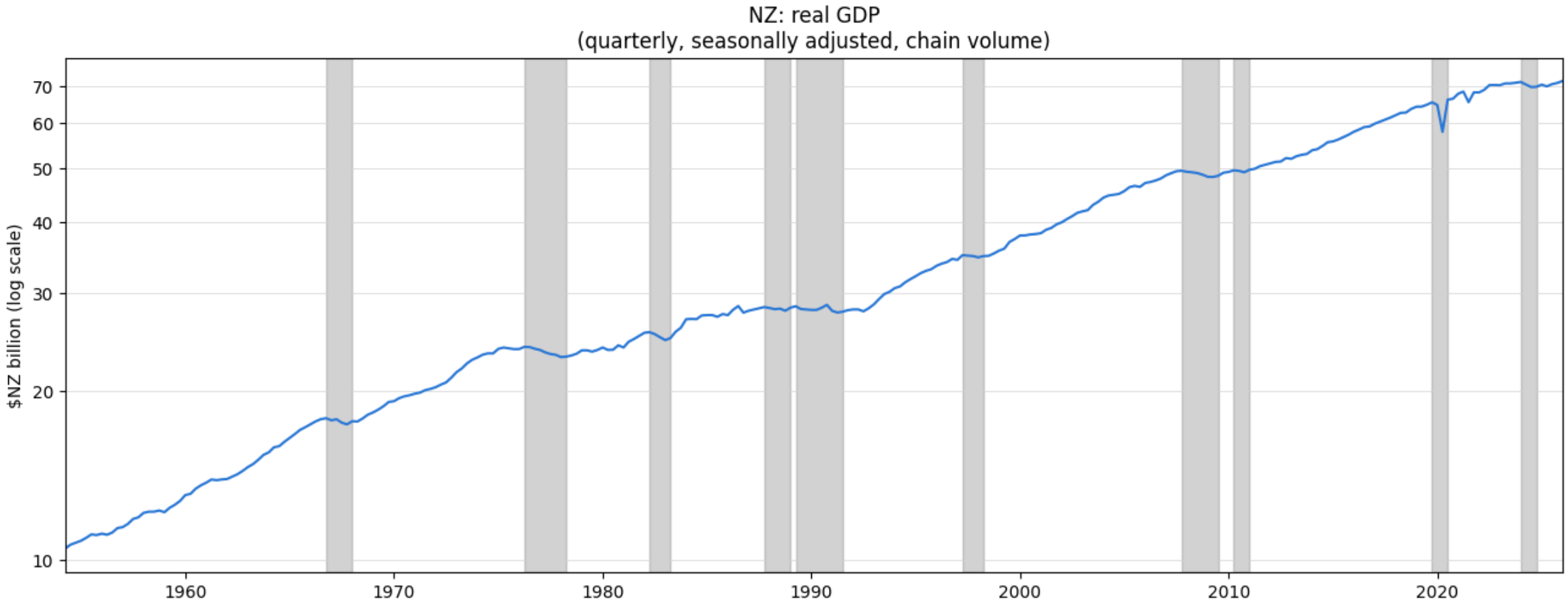
Region-specific cycles dominate the common Australasian cycle, which supports NZ keeping its own currency and monetary policy.

2020

Ask what changed

After the GFC the economy is less interest-rate sensitive, so policy must move more, and lower neutral rates have pulled down inflation and output.

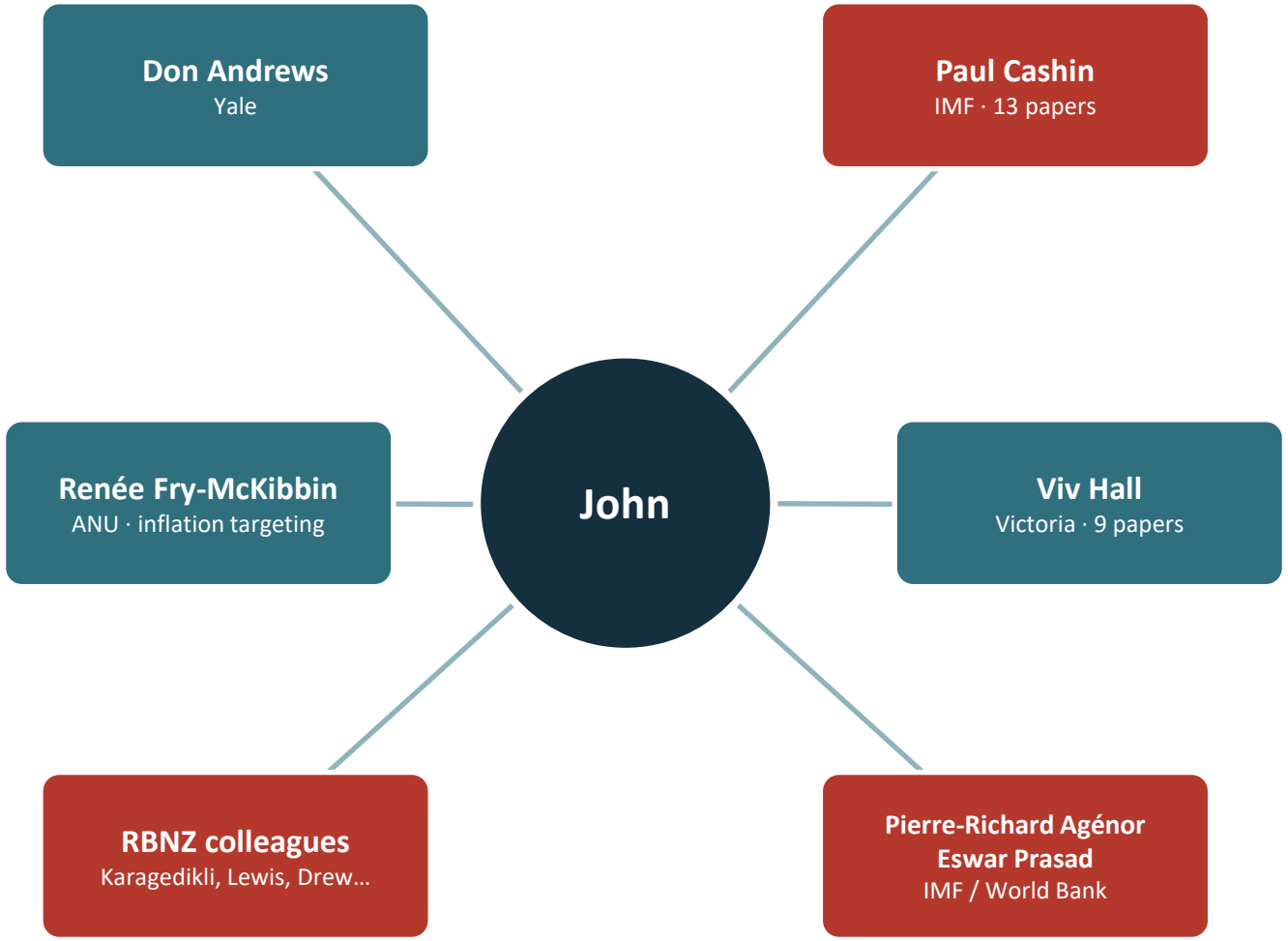
Seventy years of New Zealand recessions



Source: Hall, V. B. and McDermott, C. J. (2016) 'Recessions and recoveries in New Zealand's post-Second World War business cycles', New Zealand Economic Papers, 50(3), 261–280.

Series updated by J. McDermott, 2026

Co-authors keep the toolkit current



What a good co-author gives you

Continuity: the project survives your busy quarters

Currency: a partner closer to the academic frontier

Discipline: deadlines and referees outside your day job

Complementarity: policy questions meet new methods

Long horizons: relationships lasting 10–15 years

ACT 3

Leading research organisations

Motu and IRANZ: from doing the research to building the conditions for it



Photo: Hannah Griffin, Motu

Motu: independent research for public good

20

staff, with Senior Fellows setting their own research agendas

2000

founded as an independent, charitable research trust

2019

became Executive Director

Research initiatives led as Executive Director



Climate change



Land-use change



Carbon farming



Human rights

What changes when you lead

Senior Fellows set the research agenda. My job is to lead the organisation and get its research into New Zealand's policy debate, while securing funding and protecting independence.

IRANZ: a voice for independent research

Independent Research Association of New Zealand · Chair since 2021
As Chair, I help communicate the importance of independent research to policymakers while the science and research system is under substantial review.

28
member organisations

Members build capability across

- **Primary processing**
- **Food safety**
- **Environment**
- **Infrastructure**
- **Society**
- **Economy**
- **Health and medicine**

For doctoral graduates: independent research organisations are a real career path between universities and government.

Lessons for emerging scholars



1

Invest deeply in foundations

My PhD meant learning techniques such as cross-validation and gradient descent in depth. No one could have predicted it, but they are now central to AI models.

Foundations keep paying off: Motu is now building its own AI models.



2

Co-authors are research infrastructure

Paul Cashin and Viv Hall kept my research alive through demanding policy jobs.

Choose co-authors for complementary skills and for the long run.



3

Let policy and research shape each other

“Temporary or permanent?” and “Has the economy changed?” are policy questions and econometric questions at once.

Let the questions drive the research, and research discipline shape the policy.

Ngā mihi nui. Questions?

Pātai, kōrero, and good luck with your research

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